1 Introduction to Systems and Signals

There are 2 main concepts that we want to focus on.

1. The distinction between continuous and discrete signals
2. Fourier Series vs Fourier transforms

There exist all 4 combinations between these two -continuous time Fourier series, a discrete time Fourier series, continuous time Fourier transforms, and discrete time Fourier transforms.

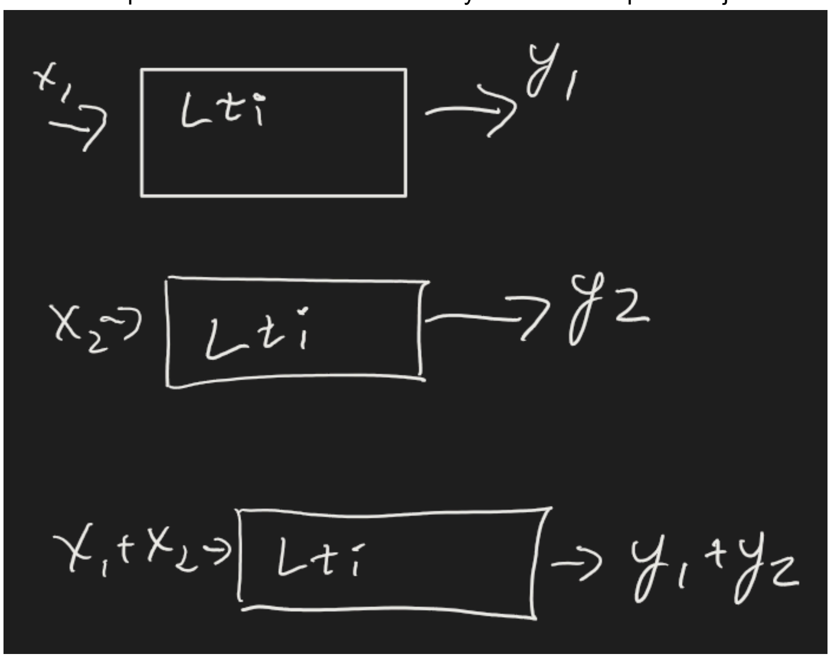
Signals

A signal is a function of 1 or more independent variables.
Examples: A pressure wave. Pressure at a position could be a function of time (speaking at a microphone).

We represent signals in the time domain or frequency domain usually (Fourier methods).

If we know the behavior of basic signals really well, then if a more complex function is a linear combination of them, then the output is a linear combination of the simpler inputs.

Take two inputs to a linear time invariant system. The outputs are just a combination of the individual function outputs.



Really, passing any functions of basic signals makes the sum of the outputs on those basic signals.

$$ax_1 + bx_2 \quad ay_1 + by_2$$

Continuous vs Discrete Time Signals

We denote continuous time by $x(t)$, whereas discrete time is $x[t]$. X is not defined if the argument t is not an integer - its a "sample number".

Think class size by graduation year - there is no 2025.2 graduating class, only at 2024 2025 and 2026 etc...

We worry about discrete time systems because we can't capture the full resolution of life - we represent and store signals in discrete manners by sampling a continuous time signal.

Systems

Systems *process* signals, they're a function or a map from input to output.

Take a spring mass system - input force and output x(t). This is a single input single output system (SISO). Similarly there are systems with multiple inputs and outputs (MIMO) (two blocks in series on springs with two output positions).

We'll look at some properties of systems.

Linearity

The scaling property applies:
$ax \to ay$.

Superpositions work:
if $x_1 \to y_1, x_2 \to y_2$ then $x_1 + x_2 \to y_1 + y_2$

Time invariance

Delaying an input by T delays generates the same output delayed by T.

If a rocket is burning fuel, it is not time invariant - mass is changing, etc. LTIs would respond to the same input in the same way no matter the time or area.

$(t) \to (t)$

$(t_0 + t) \to (t_0 + t)$

Frequency domain

Take a signal in the time domain: $x(t) = \cos(\omega_0 t)$.
In the frequency domain, this would be

$x(t) = \frac{}{2}[e^{j\omega_0 t} + e^{-j\omega_0 t}]$
$e^{j\theta} = \cos(\theta) + j\sin(\theta)$
This signal has amplitude $\frac{}{2}$ at $-\omega_0 \text{ and } \omega_0$

We can also look at sin.

$$\sin(\omega_0 t) = \frac{}{2j}(e^{j\omega_0 t} - e^{-j\omega_0 t})$$

We don't really like j on the denominator, so we write

$$\frac{1}{j} = \frac{1}{e^{j\frac{\pi}{2}}} = e^{\frac{-j\pi}{2}}$$